

WA Thenuke S Wijesinghe

Cybersecurity Internship Candidate

thenukewijesinghelin@gmail.com — +94 75 238 2969

GitHub: Thenukee — LinkedIn: Thenuke Wijesinghe — Technical Blog

Professional Summary

4th-year Computer Science undergraduate and ISC2 Certified in Cybersecurity (CC) professional with a strong foundation in threat analysis, vulnerability assessment, and digital forensics. Equipped with a solid understanding of networking and security principles, alongside hands-on experience in investigative tooling. Seeking a Cybersecurity Internship to apply analytical skills to incident handling and practical threat intelligence.

Technical Skills

- **Security Fundamentals:** Security principles, access control, authentication/authorization, security monitoring concepts, basic alert investigation
- **Networking:** TCP/IP, DNS, subnetting, routing fundamentals (CCNA level)
- **Windows Forensics & Analysis:** Windows Internals (basic), Event Logs, Prefetch, Shimcache, Amcache, Registry analysis, execution artifact correlation
- **Programming & Scripting:** Python, Java, C, SQL, PowerShell (basic)
- **Tools & Technologies:** Autopsy, IDA Freeware (basic disassembly), OSINT techniques, Docker, containerized environments, Basic Linux command-line usage
- **Documentation:** Technical writing (IEEE format), research-based reporting

Education

General Sir John Kotelawala Defence University BSc in Computer Science	2023 – 2027
University of Colombo Diploma in Human Rights	2023 – 2024
St. Joseph Vaz College, Wennappuwa Secondary Education	

Certifications

- ISC2 Certified in Cybersecurity (CC) — 2026
- Cisco CCNA: Introduction to Networks — 2023
- Cisco NDG Linux Unhatched — 2026

Projects

- **Behavioral Keystroke Timing Analyzer** *.NET 8, WinForms (Ongoing)* — Building a Windows app to capture keystroke timing patterns for behavioral authentication research.
[View Project](#)
- **Simulated SIEM Log Analysis Pipeline** *Python* — Built a pipeline to analyze Windows Event Logs and detect suspicious activity using rule-based alerting to simulate SIEM workflows.
[View Project](#)
- **IoT-Based Emergency Alert and Evidence Collection System** *ESP32, Arduino* — Developed a real-time alert system integrating GPS and GSM to transmit location and sensor data for emergency response.
[View Project](#)
- **OSINT Data Collection and Link Analysis Platform** *Python, Flask, Docker* — Built a containerized platform to collect, correlate, and visualize intelligence data using modular

scraping and graph analysis. Developed for CID Sri Lanka. [View Project](#)

- **Local TCP Port Scanner (Defensive Security Tool)** *Python* — Developed a TCP port scanner to identify exposed services, simulating reconnaissance for defensive lab use.
[View Project](#)
- **File Integrity Verification Tool (Digital Forensics Utility)** *Python* — Built a lightweight CLI tool using MD5/SHA-256 hashing to verify evidence integrity in forensic workflows.
[View Project](#)
- **Password Strength and Generation Utility** *Python* — Built a tool to generate strong randomized passwords and promote secure credential practices.
[View Project](#)

Research Experience

Design and Evaluation of a Proactive Digital Forensics Triage Framework for Stealthy Malware Detection on Windows 11 24H2 (Final Year Research Project, Ongoing)

- Designed and prototyped a hybrid proactive digital forensics framework that augments existing EDR/AV solutions through configurable artifact baselining and triggered memory forensics.
- Developed a three-layer architecture combining periodic differential analysis of key Windows artifacts (registry, processes, KMDs, Services) with full physical memory acquisition.
- Implemented triggered and optional periodic memory dump mechanisms with resource throttling, followed by automated offline memory forensics for IOC extraction and visibility.
- Aimed at bridging the EDR investigation gap for stealthy, fileless, and in-memory malware.

Leadership & Experience

Co-Founder, CLAPTAC

- Led development of web applications and POS systems, managing the full development life-cycle.
- Conducted requirement analysis and translated business needs into technical solutions.

Awards & Achievements

- **Byte-Blitz: Flash Code Frenzy 2024** — 1st place (competitive programming).
- **WSO2 Ballerina Competition 2025** — Top 10 teams (national level).
- **CICRA 2025 CTF Challenge** — Top 10 participants (cybersecurity CTF).
- **Code with WIE 2025 (IEEE WIE Sri Lanka)** — Top 10 teams.
- **Mini Hackathon 2025 (University of Colombo)** — Finalist.
- **Cypher 3.0 2025 (KDU IEEE WIE)** — Finalist.
- **IEEE Xtreme 18.0** — Top 25 Sri Lanka (global 24-hour contest).

Referees

Dr. Kaneeka Vidanage

HOD – Department of Computer Science,
General Sir John Kotelawala Defence University
Email: vidanage_bvki@kdu.ac.lk
Phone: 071 3543786

Mrs. D. V. Dharani Abeysinghe

Lecturer - General Sir John Kotelawala Defence University
Email: abeysinghe.dvds@kdu.ac.lk
Phone: 076 678 77887